25900270

MAHATMA GANDHI UNIVERSITY, KOTTAYAM MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCBTG201, MG3DSCBBT201

CELL BIOLOGY AND GENETICS

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

One word or sentence

Answer any 10 questions-Each question carries 2 marks

1. What are the different types of Sex linked Inheritance?[U] / [CO3]

   Types of Sex linked Inheritance-X linked, Y linked, XY- linked – 2 marks
2. What is macroautophagy? [U] / [CO2]

   Macroautophagy- Definition-2 marks
3. Define Mutation. [U] / [CO3]

   Mutation- definition- 2 marks
4. Name two functions of cholesterol in the plasma membrane. [U] / [CO1]

   Any two functions – 2 marks
5. Name the scientist who proposed the inheritance of A and B blood groups?. [K] / [CO3]

   Karl Landsteiner- 2 marks
6. Name the enzyme which is present in peroxisomes to break down hydrogen peroxide. [K] / [CO2]

   Catalase- 2 marks
7. Name the organelle which plays a key role in forming primary lysosomes. [K] / [CO2]

   Golgi apparatus-2 marks
8. What is a karyotype? [K] / [CO3]

   Definition- 2 marks
9. What does non-random mating cause in a population? [K] / [CO3]

   Influences genetic diversity/ lead to a decrease in overall genetic variation/reduction in effective population size/ increase genetic drift- 2marks
10. Which type of transport involves the movement of substances from high to low concentration without energy?[K] / [CO1]

    Passive transport-2 marks

11. Which type of Sex determination system is present in Drosophila Melanogaster? [K] / [CO3]

    X/A ratio- 2 marks

12. What is euchromatin? [U] / [CO2]

    Definition- 2 marks [2x10 = 20]

Part B

Short Essay

Answer any 5 questions-Each question carries 4 marks

13. Mention the characteristics of prokaryotic and eukaryotic cells [U] / [CO1]

    Characteristics of prokaryotic and eukaryotic cells- minimum 4 points (each)-4marks

14. Differentiate sex-linked traits and autosomal trait. [U] / [CO3]

    Sex-linked traits -definition, example- 1 mark, explanation-1 mark, autosomal trait - definition, example- 1 mark, example- 1 mark.

15. Write a note on the structure and functions of glycocalyx.[U] / [CO1]

    Structure- 2 marks, functions- 2 marks

16. Discuss the diverse roles played by integrins in ECM. [K] / [CO1]

    Integrins in ECM-introduction- 2 marks, Role-(minimum four points)- 2 marks

17. Explain why mitochondria is called the powerhouse of the cell. [U] / [CO2]

    Mitochondria- introduction- 1mark, powerhouse of the cell-Explanation- 3 marks

18. Describe the different stages of crossing over. [U] / [CO3]

    Crossing over-Definition- 1 mark, stages-2 marks, diagram- 1 mark.

19. Briefly explain Mendalian laws. [U] / [CO3]

    Mendalian laws- explain- 4 marks

[4x5 = 20]

Part C

Essay type questions

Answer any 1 questions-Each question carries 10 marks

20. Describe the structure and functions of endoplasmic reticulum in eukaryotic cells.[U] / [CO2]

    Endoplasmic reticulum-Structure with diagram- 2 marks, Types- 2 marks, Functions- 6 marks.

21. Describe in detail the extra-chromosomal inheritance shown by mitochondria with suitable examples. [U] / [CO3]

    Extra-chromosomal inheritance-definition-2 marks, Criterias- 2 marks, 2 examples-explain-6 marks.

[10x1 = 10]